SMRTTECH 4ID3 SCHEDULE OF LABS

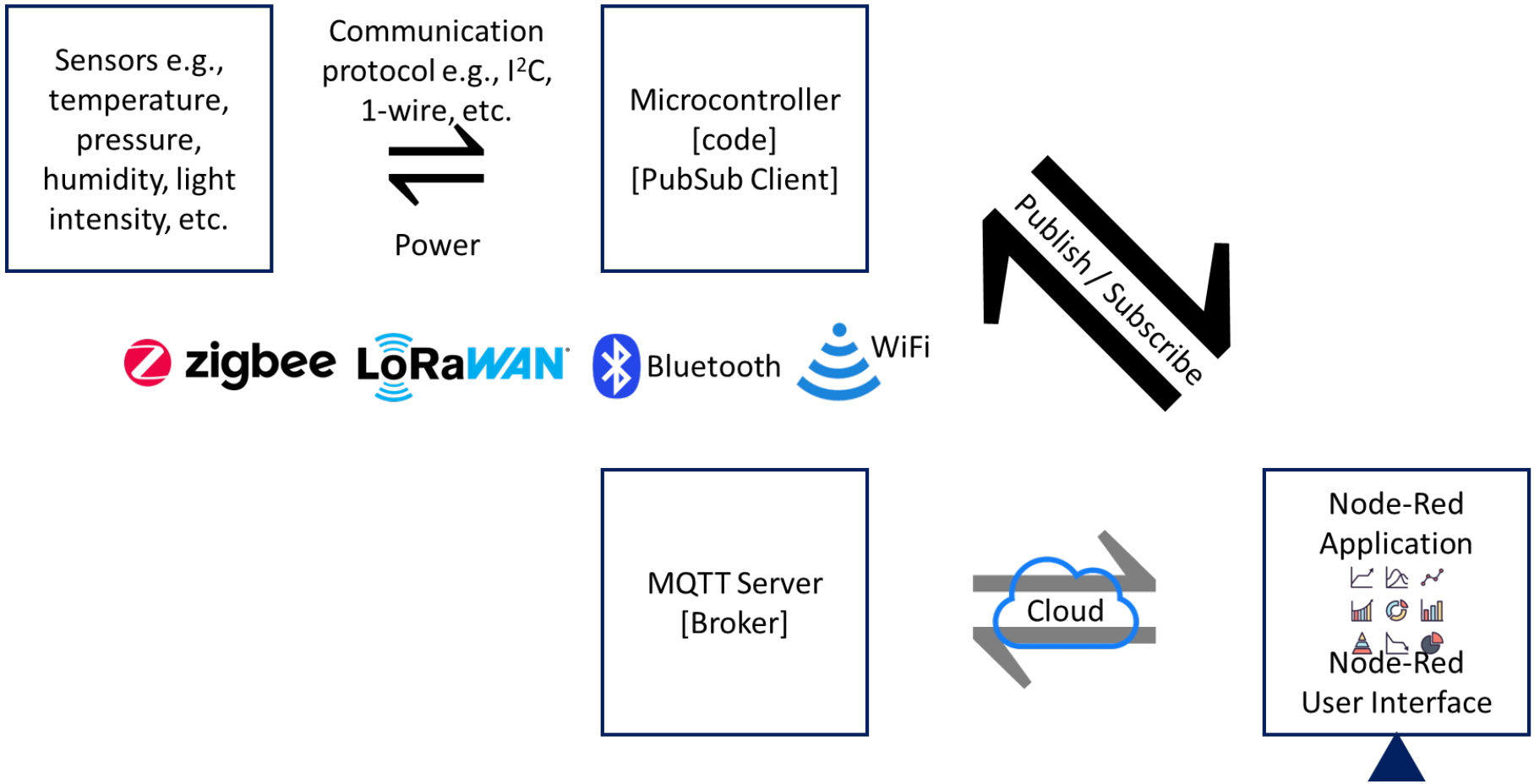


Figure: 01 A Typical Internet of Things Setup with Focus on varying Access Layer Technologies; communication enabled via MQTT Application Protocol

Lab #	Lab Title	Purpose/Objective	Learning Outcome
WEEK 1 Lab 01 Week of Mon. 05 Jan.	Introduction to Lab Work and Projects	An understanding of how labs and projects work and are related, along with initiating community-based research for project challenge identification, and forming project teams.	Students learn how to research for and draft a community-based project challenge using a theme-based approach to narrow down to a proof-of-concept project. Additionally, students form their project teams.
WEEK 2 Lab 02 Week of Mon. 12 Jan.	Introduction to IoT WiFi	A practice-based understanding on the "things" of the Internet of Things, and the "Internet" of the Internet of Things, including temperature, pressure, humidity, and light intensity sensors, connected to a microcontroller (with Publication-Subscription client), communicating over **802.11ah WiFi** to an MQTT server on cloud, corresponding with Node-Red application, showcasing outcome in a Node-Red user interface.	Students learn appreciation for Node-Red, Mosquitto, MQTT, various common sensors, ESP 8266 microcontroller, Arduino IDE, and its specific uses for a typical IoT Setup (**Figure 01**). Additionally, students learn to install programs on their computers and smart phones. Students learn about WiFi as an Access Technology, and appreciate the criteria for choosing WiFi for a given application.
WEEK 3 Lab 03 Week of Mon. 19 Jan.	Understanding coding Bluetooth	Thorough understanding of the code in Lab 01 including variations that may follow A practice-based understanding on the "things" of the Internet of Things, and the "Internet" of the Internet of Things, including temperature, pressure, humidity, and light intensity sensors, connected to a microcontroller (with Publication-Subscription client), communicating over **Bluetooth** to an MQTT server on cloud, corresponding with Node-Red application, showcasing outcome in a Node-Red user interface.	Students lean, line-by-line, explanation of a generic code, enabling them to edit code as needed for lab and/or project work. Students learn about Bluetooth as an Access Technology, and appreciate the criteria for choosing Bluetooth for a given application.
WEEK 4 Lab 04 Week of Mon. 26 Jan.	Project pitches and resources needed	Project pitches from each student team A collective identification of technical resources needed; bill of materials; budget	Students learn to pitch their project proposals, including identifying project requirements/needs for a demonstratable presentation.
WEEK 5 Lab 05 Week of Mon. 02 Feb.	LoRa Addressing concerns with Project Challenge	A practice-based understanding on the "things" of the Internet of Things, and the "Internet" of the Internet of Things, including temperature, pressure, humidity, and light intensity sensors, connected to a microcontroller (with Publication-Subscription client), communicating over **LoRa** to an MQTT server on cloud, corresponding with Node-Red application, showcasing outcome in a Node-Red user interface. Finalizing technical resources needed; bill of materials; budget	Students learn about LoRa as an Access Technology, and appreciate the criteria for choosing LoRa for a given application. Students learn to pitch their project proposals, including identifying project requirements/needs for a demonstratable presentation.

WEEK 6 Lab 06 Week of Mon. 09 Feb.	LoRaWAN Addressing concerns with Project Challenge	A practice-based understanding on the "things" of the Internet of Things, and the "Internet" of the Internet of Things, including temperature, pressure, humidity, and light intensity sensors, connected to a microcontroller (with Publication-Subscription client), communicating over **LoRaWAN** to an MQTT server on cloud, corresponding with Node-Red application, showcasing outcome in a Node-Red user interface. Finalizing technical resources needed; bill of materials; budget	Students learn about LoRaWAN as an Access Technology, and appreciate the criteria for choosing LoRaWAN for a given application. Students learn to pitch their project proposals, including identifying project requirements/needs for a demonstratable presentation.
WEEK 7 Week of Mon. 16 Feb.	Mid-term recess(es)		
WEEK 8 Lab 07 Week of Mon. 23 Feb.	ZigBee Projects	A practice-based understanding on the "things" of the Internet of Things, and the "Internet" of the Internet of Things, including temperature, pressure, humidity, and light intensity sensors, connected to a microcontroller (with Publication-Subscription client), communicating over **ZigBee** to an MQTT server on cloud, corresponding with Node-Red application, showcasing outcome in a Node-Red user interface. Begin project work in labs	Students learn about ZigBee as an Access Technology, and appreciate the criteria for choosing ZigBee for a given application. Students initiate project work in labs, learning real world problems of troubleshooting in IoT-based projects.
WEEK 9 Lab 08 Week of Mon. 02 Mar.	Projects	Continuing project work in labs Troubleshooting project work in labs Catering to individual project needs	Students continue project work in labs, learning real world problems of troubleshooting in IoT-based projects, including troubleshooting specifics based on chosen project.
WEEK 10 Lab 09 Week of Mon. 09 Mar.	Projects	Continuing project work in labs Troubleshooting project work in labs Catering to individual project needs	Students continue project work in labs, learning real world problems of troubleshooting in IoT-based projects, including troubleshooting specifics based on chosen project.
WEEK 11 Lab 10 Week of Mon. 16 Mar.	Projects	Continuing project work in labs Troubleshooting project work in labs Catering to individual project needs	Students continue project work in labs, learning real world problems of troubleshooting in IoT-based projects, including troubleshooting specifics based on chosen project.
WEEK 12 Lab 11 Week of Mon. 23 Mar.	Projects Project Presentations Dry run	Continuing project work in labs Troubleshooting project work in labs Catering to individual project needs Final Project Presentation Dry run	Students continue project work in labs, learning real world problems of troubleshooting in IoT-based projects, including troubleshooting specifics based on chosen project.

			Students learn to rehearse for final presentations, including showcasing accomplished work, and acknowledging need for continuing work.
Week 13 Lab 12 Week of Mon. 30 Mar.	Project Presentations	Final Project Presentations	Students learn to present high level objectives of the project, starting with community pain points, and proposed proof-of-concept; presenting work accomplished as part of proof-of-concept; demo of functional project; articulating future work needed.
Week 14 Lab 13 Week of Mon. 06 Apr.	Project Presentations	Final Project Presentations	Students learn to present high level objectives of the project, starting with community pain points, and proposed proof-of-concept; presenting work accomplished as part of proof-of-concept; demo of functional project; articulating future work needed.